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11

DSMC PROCEDURES IN A HOMOGENEOUS GAS

11.1 Collision sampling techniques

The DSMC procedures for the simulation of a representative set of intermolecular collisions are best derived and demonstrated in the context of a homogeneous or 'zero-dimensional' gas. However, in order to make the procedures directly useful as subroutines in the later programs, the homogeneous gas is divided into a one-dimensional structure of cells and sub-cells. This also permits the testing of the procedures for any effects due to the number of molecules per cell or sub-cell. The procedures for the establishment of the correct collision rate are based on the cells, while individual collision pairs are chosen from the sub-cells.

The probability of a collision between two molecules in a homogeneous gas is proportional to the product of their relative speed c_r and total collision cross-section σ_T. Eqn (1.11) for the non-equilibrium collision rate in a homogeneous gas could be used to establish the number $N_c \Delta t$ of collisions in each cell at each time step of duration Δt, and this number could then be computed. The mean value of the product of c_r and σ_T is calculated for each cell, and the maximum value could also be recorded. The collision pairs could then be chosen by the acceptance–rejection method, with the probability of a particular pair being given by the ratio of their product of c_r and σ_T to the maximum product. However, this procedure would have a computation time proportional to the square of the total number of molecules in the cell.

The so-called 'time-counter' or TC method was therefore introduced (see, for example, Bird 1976) in order to obtain a computation time directly proportional to the number of molecules. This involved the calculation of representative collisions and, at each collision, a time parameter in each cell was advanced by the amount appropriate to that collision. The process was repeated until the cell time caught up with the flow time. This process had the optimal computational efficiency, and it was shown analytically that it led to the correct non-equilibrium collision rate. However, in extreme non-equilibrium situations such as the leading edge of extremely strong shock waves, there were problems due to correlations between the probability and the spacing of the collisions in time. The occasional acceptance of a collision pair with a small collision probability could result in the cell time being advanced by such a large interval that no other collisions could be computed for a number of time steps. A number of

alternative schemes have been proposed and some of these (for example, Koura 1986) involve only a small time penalty. However, the programs in this book employ a scheme called the NTC method (Bird 1989) which completely avoids the problems with the TC method without any loss of computational efficiency. It has the further advantage that the number of pair selections is determined in advance of the entry to the collision coding, rather than within this coding when the cell time reaches the flow time. This allows easy vectorization of the collision procedure.

Consider a DSMC cell of volume V_C in which each simulated molecule represents F_N real molecules. The probability P of collision between two simulated molecules over the time interval Δt is equal to the ratio of the volume swept out by their total cross-section moving at the relative speed between them to the volume of the cell, i.e.

$$P = F_N \sigma_T c_r \Delta t / V_C . \tag{11.1}$$

The relative speed varies with the choice of the collision pair and the total cross-section is generally a function of the relative speed, but the other quantities in eqn (11.1) are independent of this choice. The average number of real molecules in the cell is nV_C and the average number of simulated molecules is $N = nV_C/F_N$, where n is the number density in the real gas. The full set of collisions could be calculated by selecting, in turn, all $N(N-1)/2$ pairs in the cell and by computing the collisions with probability P. This method has been used in DSMC simulations, but is inefficient because P is generally a very small quantity and the number of choices is very nearly proportional to the square of the number of molecules. Also, since F_N is an extremely large number, the number of choices should be $N^2/2$, and this error becomes significant for small values of N. The procedure can be made more efficient and the second difficulty removed if only a fraction of the pairs are included and the resultant probability increased by dividing eqn (11.1) by this fraction. Maximum efficiency is achieved if the fraction is such that the maximum probability becomes unity. The fraction is then given by

$$P_{max} = F_N (\sigma_T c_r)_{max} \Delta t / V_C , \tag{11.2}$$

and the number of pair selections per time step is obtained by multiplying this equation by $N^2/2$. However, in most cases, N is a fluctuating quantity and, because the mean of the square differs from the square of the mean, N^2 should be replaced by the product of the instantaneous value and a time or ensemble averaged value. Therefore, the NTC method is that

$$\tfrac{1}{2} N \bar{N} F_N (\sigma_T c_r)_{max} \Delta t / V_C \tag{11.3}$$

pairs are *selected* from the cell at the time step, and the collision is computed with probability

$$\frac{\sigma_T c_r}{(\sigma_T c_r)_{max}} . \tag{11.4}$$

Note that, if N is doubled through halving F_N, the number of pair selections also doubles, so that the computing time is linear with N. The parameter $(\sigma_T c_r)_{max}$ should be stored for each cell and is best set initially to a reasonably large value, but with provision for it to be automatically updated if a larger value is encountered during the sampling. Note that this parameter occurs in the numerator of eqn (11.3) and the denominator of eqn (11.4), so that the rate is not affected by its exact value. Within the statistical scatter, the NTC method yields the exact collision rate in both simple gases and gas mixtures, and under either equilibrium or non-equilibrium conditions.

In the case of a gas mixture, the procedures are species independent and the NTC method is best applied to all the molecules as a single group. However, a practical problem arises when there is a very large variation in the molecular masses. This is because the collisions that involve the light molecules will have high values of the relative speed and, since these will set the value of $(\sigma_T c_r)_{max}$, the acceptance rate for the heavy molecules will be low, and the overall selection process will be inefficient. The method can be applied separately to the species and, for a collision between a species p and a species q molecule, eqns (11.3) and (11.4) can be written

$$\tfrac{1}{2} N_p \bar{N}_q F_N \{(\sigma_T c_r)_{max}\}_{pq} \Delta t / V_C \tag{11.5}$$

and

$$\frac{\sigma_T c_r}{\{(\sigma_T c_r)_{max}\}_{pq}}, \tag{11.6}$$

respectively. This causes a storage problem when there are a large number of species in that two variables, that are doubly subscripted with respect to the total number of species, must be defined for each sub-cell. The solution is to split the molecules into several groups and to regard p and q as defining groups of species rather than a single species.

11.2 Collision test programs

A FORTRAN 77 program, called DSMC0.FOR, has been written to test the NTC procedure in a homogeneous gas mixture. The number of simulated molecules per cell is an important factor in assessing the performance of a collision scheme so that, although the gas is uniform, it is divided into a set of cells and sub-cells. The geometry is a one-dimensional gas of unit cross-section between two plane specularly reflecting walls that are normal to the x-axis. The overall structure of the demonstration programs is illustrated by the listing of program DSMC0S.FOR in Appendix G.

The only variables that are at the disposal of the user are those in the PARAMETER statement that set the dimensions of the subscripted variables, and those in subroutine DATA0 that set the data for a particular run of the program. The subroutine that sets the data will always be listed as the final subroutine in a program. The main program contains an inner

loop over the number NIS of time steps between flow samples and a second loop over the number NSP of samples between the updating of the restart file DSMC0.RES and the output file DSMC0.OUT. The program stops after NPT update cycles have been completed. Apart from these loops, the main program calls subroutines that are, as far as possible, standardized for inclusion in other programs.

The first subroutine is named INIT0, and is called whenever a new run is started by entering '1' when the program requests a code for a continuing or new calculation. This subroutine sets the physical constants and then calls the data subroutine. The data is limited, in this case, to information on the individual species, and this is converted to the fuller information on all classes of collision. The flowfield is divided into MNC cells of uniform width CW, and each of these is divided into NSC equal width sub-cells. Finally, the required random locations and velocity components of the initial state of the gas are set. Because the flow is one-dimensional, only the x-coordinate is stored for each molecule. All positions are equally likely on the plane normal to the x-axis. Note that all three velocity components are stored, and the collisions are computed as three-dimensional events. The second subroutine SAMPI0 initialises the sampling variables.

Subroutine MOVE0 moves the molecules through distances appropriate to the time interval DTM. Apart from the specular reflections from the end walls at $x = \mathrm{XF}$ and $x = \mathrm{XR}$, the procedures are trivial. Specular reflection reverses the velocity component normal to the surface, while those parallel to the surface are unchanged. The final position of a molecule that strikes a surface normal to the x-axis at x_c and would have moved to x' beyond it is given by

$$x_c - x = x' - x_c ,$$

or

$$x = 2x_c - x' . \tag{11.7}$$

Subroutine INDEX sorts the molecule numbers in the one-dimensional array IR into the order of the groups and, within a group into the order of the cells. Finally, within the cell, it arranges them in the order of the sub-cells. This sorting procedure employs multi-level linked lists and the computation time is directly proportional to the total number of molecules. The cross-reference array permits the efficient selection of pairs of molecules from the same group of species and from the same, or the nearest possible, sub-cell as potential collision pairs.

The NTC method is employed in subroutine COLLM to determine the appropriate set of collisions. This includes subroutines SELECT for the selection of representative collision pairs and ELASTIC for the computation of the collision. The comment statements refer to the equations on which the coding is based. The selection of the post-collision components of the relative velocity in the centre of mass frame of reference in subroutine ELASTIC requires further discussion. The deflection angle of $\boldsymbol{c}_r$ for the VSS model in this frame of reference is given by eqn (2.36), i.e.

$$\cos(\chi/2) = (b/d)^{1/\alpha},$$

where the diameter d is the effective diameter for collisions between the two molecules. Therefore,

$$\cos\chi = 2\{(b/d)^2\}^{1/\alpha} - 1,$$

and the square of the ratio of the impact parameter b to d is uniformly distributed between 0 and 1. This distribution is that of the random fraction itself, so that the selection rule can be written

$$\cos\chi = 2R_f^{1/\alpha} - 1. \tag{11.8}$$

For the VHS model, $\alpha = 1$, and all directions are equally probable for the post-collision direction of the relative velocity. Advantage may be taken of this to replace χ in eqn (11.6) by an elevation angle θ, and an azimuth angle ϕ may be chosen from a uniform distribution between 0 and 2π. The three post-collision components of the relative velocity are then given by multiplying the relative speed by $\cos\theta$, $\sin\theta\cos\phi$, and $\sin\theta\sin\phi$. The VSS model requires the computation of the post-collision relative velocity components through the far more complex relations of eqn (2.22). Should the scattering parameter α be very close to unity, the VHS logic is applied in place of the VSS logic.

Both the indexing subroutine INDEX and the collision subroutine COLLM have been greatly complicated by the inclusion of the molecular groups that are, as explained above, required only in the case of gas mixtures with large mass differences. As well as making the programs more difficult to read, this complication involves a computational overhead of several percent. However, while this program is only for the testing of the collision procedures, the aim is to develop general subroutines that are capable of handling all applications of the DSMC method. The simpler procedures for a simple gas are illustrated in the program DSMC0S.FOR.

The flow properties in each cell are sampled within the subroutine SAMPLE, and the output file DSMC1.OUT is updated in the subroutine OUT0. The latter subroutine employs the definitions of the macroscopic properties that have been given in the earlier chapters, and the equations are identified in the comments. Double precision is used for variables such as the total number of collisions that are expected to be advanced beyond the limit of single precision arithmetic (see Appendix E). The operation of the indexing and collision selection procedures was tested through sampling the mean separation of the molecules in collisions. With an average of 2.5 molecules in each sub-cell and with all the molecular species in a single group, the mean spacing was found to be just under 40% of the sub-cell width. This means that the sampled collisions are close to the ideal of all collisions being between nearest neighbours. In addition, the acceptance rate of the collision pairs in eqn (11.4) was sampled. The maximum value of the product of the relative speed and cross-section was initially set to the product of the reference cross-section of the most prevalent combination of

these species and 300 ms^{-1}, which approximates to the most probable speed of these molecules at the initial temperature. For the test case data that is described in the following paragraph, the initial acceptance rate was approximately 55%, and this declined to around 45% as the maximum value increased with the occurrence of unlikely collisions with very fast relative speeds. As explained in the preceding section, the collision rate is correct as long as the maximum value is reasonably high, and it is not affected by the gradual growth of the maximum value.

As noted earlier, the data that affects the dimensions of the arrays is set in the PARAMETER statement. The test case employs 1000 molecules (MNM=1000), 50 cells (MNC=50) with eight sub-cells per cell (MNSC=400), and five species (MNSP=5) in a single group (MNSG=1). The remaining variables are set in the DATA0 subroutine. The initial gas has a number density of 10^{20} m^{-3} at a temperature of 300 K. The number fractions of the molecular species range from 0.6 for the most prevalent species to 0.02 for the least prevalent. The flowfield extends in the x-direction from the origin to $x = 1$ m, and its cross-section is assumed to be 1 m^2. Therefore, in order to obtain 1000 molecules, the number of real molecules that are represented by each simulated molecule is set to 10^{17}. The cell width is then 0.02 m and, while this is of the order of the equilibrium mean free path, there are no gradients in the flow and the sub-cell width is small in comparison with the mean free path. A typical molecular velocity component is 200 ms^{-1}, so that a time step of 0.000025 s leads to a typical displacement in the x-direction of 0.005 m. This is 25% of the cell width and this value has been chosen as the time step. The flow is sampled after each fourth time step (this should lead to only a small degree of correlation between successive samples) and the files are updated at intervals of 40 samples. The program stops after 500 file updates. Because this is simply a test of the procedures, the molecular masses and diameters have been set to nominal values that are of the order of those in real gases. In addition, many of the molecular properties that would normally be set to separate values in the data subroutine have been set to common values in the initialization subroutine. The temperature index of the viscosity coefficient is set to 0.75 for all interactions and the VHS model has been selected through the setting of the scattering parameter to unity.

Because of the relatively small sample of 1000 simulated molecules in this test case, and the fact that the initial velocity components of each molecule are set independently, the overall gas temperature differs from the nominal value. The overall temperature was found to vary by one or two degrees from the nominal value of 300 K. Also, although the gas is nominally at rest, there is a finite velocity due to the scatter. For this sample, the individual velocity components were found to be of the order of five to 10 ms^{-1}.

The total number of collision events counted by the variable NCOL was 17,214,867 and, for each of these, the appropriate values of the doubly subscripted variable COL were advanced by one. The final state of NCOL from a run of DSMC0.FOR with the listed data is shown in Table 11.1.

Table 11.1 Sample size of collisions

Species	1	2	3	4	5
1	17,590,774	6,923,654	3,101,569	2,679,354	712,407
2	6,923,654	2,683,570	1,222,127	1,053,081	277,859
3	3,101,569	1,222,127	504,620	430,004	125,206
4	2,679,354	1,053,081	430,004	357,630	107,509
5	712,407	277,859	125,206	107,509	27,600

Table 11.2 Ratio of the calculated to theoretical collision rates.

Species	1	2	3	4	5
1	0.999384	1.000546	1.001161	1.000897	0.997228
2	1.000546	0.996645	1.002060	1.001812	1.001123
3	1.001161	1.002060	0.993330	0.997551	1.004428
4	1.000897	1.001812	0.997551	0.983847	1.003631
5	0.997228	1.001123	1.004428	1.003631	0.966407

This sampled collision rate is compared in Table 11.2 with the exact analytical result of eqn (4.78) for an equilibrium gas mixture. Apart from the rate for collisions between the trace species (number 5), the collision rates are in excellent agreement with the theoretical values. That for the 5-5 collision is low by about five standard deviations, so that the discrepancy is significant. The average number of molecules per sub-cell is 2.5 so that, if we consider a sub-cell containing a species 5 molecule, the probability that there is another type 5 molecule in the sub-cell is less than one in thirty. Given the adverse statistics for the trace molecules, and the fact that the molecular species was ignored in the selection of potential collision pairs, the results are surprisingly good.

The run was repeated with each species specified as a separate group. Each of the 25 classes of collision was then treated separately with regard to the selection of collision pairs and the establishment of the collision rate. The result for the collision rate is shown in Table 11.3, and the rate for the class 5-5 collisions is now in agreement with the theory. However, since the species of the collision pair is now specified, the selection is generally forced beyond the sub-cell, and the mean spacing between collision pairs (averaged over all species) increased from 0.00098 to 0.00247, which approximates the width of the sub-cell. This increase is undesirable and, in general, it would be preferable to accept the slightly low collision rate for self collisions between the trace species.

Table 11.3 The ratio of collision rate to the theoretical value when each species forms a separate group.

Species	1	2	3	4	5
1	1.001099	1.000952	1.000459	1.001607	1.001462
2	1.000952	1.002584	0.999471	0.999052	0.997745
3	1.000459	0.999471	1.004441	0.998531	0.996351
4	1.001607	0.999052	0.998531	1.002897	0.993467
5	1.001462	0.997745	0.996351	0.993467	1.004225

The specification of separate collision groups for each species also leads to an increase in the computation time, even though the acceptance ratio for collision pairs is higher. The greatest concern with this acceptance ratio is that the maximum value of the product $(\sigma_T c_r)_{max}$ of the relative speed and collision cross-section may increase during a very long run to such an extent that the selection process becomes unduly inefficient. Therefore, for the case with all molecules in a single group, the mean value of this product over all cells was sampled. The minimum and maximum values in any cell were also recorded and the results are shown in Fig. 11.1. The initial value of the product was set to the excessively low value of 1.155×10^{-16} m^3s^{-1}, but it increased during the first few collisions to values that allowed the correct collision rate to be established. The important conclusion from this test is that, even in a long run, the rare very fast collisions do not increase $(\sigma_T c_r)_{max}$ to the stage where the selection procedure becomes excessively inefficient. The collision rate would be unaffected and the program would be slightly more efficient if the values of $(\sigma_T c_r)_{max}$ were reset in steady flows to their value at (say) one million collisions at each successive million collisions. This is not done in the demonstration programs, but it is recommended that the collision pair acceptance ratio should be included in the output and be monitored.

In addition to checking that the procedures produce satisfactory results when the computational parameters, such as the number of molecules per cell, have their recommended values, the way in which the simulation breaks down for unreasonable values of these parameters should be investigated. The earlier test cases were therefore rerun with the variable FNUM increased by a factor of ten, so that the number of simulated molecules was reduced to 100. The average number of molecules per cell was then only two, and there was only one molecule to every four sub-cells. Table 11.4, for the case in which all species are in one group, shows that the error in the rate for the 5-5 collisions between trace molecules is now very significant, and there are discrepancies in the rates for the other collisions between the less prevalent species. The mean separation between collision pairs increased to 0.00474 m which is five times worse than the case with an adequate number of molecules per cell.

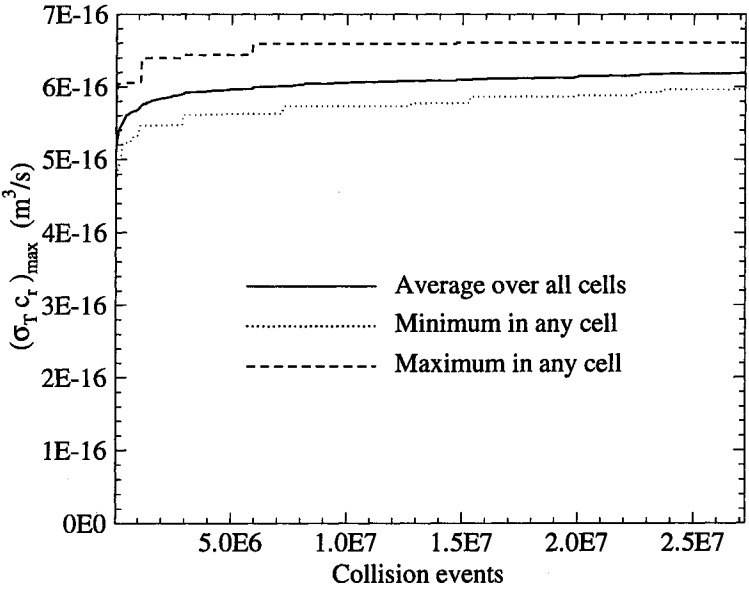


Fig. 11.1 The sampled values of the maximum value of the product of the relative speed and the collision cross-section.

When all species were placed in separate groups, Table 11.5 shows that all collision rates were about two to three percent higher than the theoretical rates. The mean spacing between collision pairs increased to 0.00614 m. Molecular groups can be used to obtain more accurate rates for collisions between trace species, but this is at the cost of increased separations between collision pairs, and is not necessary if the average number of molecules per cell and sub-cell meet the recommended minimum values.

Table 11.4 Comparison with theoretical collision rates with only two molecules per cell and all species in a single group.

Species	1	2	3	4	5
1	0.997301	1.016905	1.036046	1.043018	1.019806
2	1.016905	0.968886	1.033997	1.046500	1.021853
3	1.036046	1.033997	0.952629	1.062859	1.053322
4	1.043018	1.046500	1.062859	0.936944	1.071641
5	1.019806	1.021853	1.053322	1.071641	0.503295

Table 11.5 Comparison with theoretical collision rate with only two molecules per cell and all molecular species in separate groups.

Species	1	2	3	4	5
1	1.025179	1.025466	1.025064	1.025147	1.028721
2	1.025466	1.034196	1.022251	1.022558	1.029195
3	1.025064	1.022251	1.030747	1.021415	1.022737
4	1.025147	1.022558	1.021415	1.025757	1.022754
5	1.028721	1.029195	1.022737	1.022754	1.018747

To illustrate the circumstances in which molecular groups should be used, a calculation was made for a mixture of 10% helium and 90% xenon, by number. The overall number density, the temperature, the number of simulated molecules, and the numbers of cells and sub-cells were the same as in the test case. With a single molecular group, the overall acceptance rate of collision pairs was 26%, but this increased to 46% when each species was regarded as a single group. This led to a 25% reduction in the computing time, but the mean spacing between collision pairs increased by 60%. A large mass ratio such as this is comparatively rare and, even though the reduction in computing time is significant, the case for the inclusion of molecular groups in the standard DSMC routines is marginal. The case for their inclusion does become overwhelming when electrons are present, and this was the critical consideration.

As noted earlier, it is debatable whether there should be alternative routines and programs when dealing with simple gases rather than gas mixtures. A simple gas version of DSMC0.FOR was therefore produced in order to assess the penalty due to the redundant code when the mixture code is applied to a simple gas. This program is listed in Appendix G as DSMC0S.FOR. For the data in subroutine DATA0S, the simple gas program was only 11% faster than the mixture program running the corresponding data. Comparisons of computer time can be compiler dependent, particularly in cases such as this where the redundant loops over molecular species could be removed by an optimizing compiler, but this result indicates that it is not worth maintaining separate programs for simple gases and gas mixtures. The remaining programs will be for the more general case.

The value of the VSS scattering index data in DSMC0S was changed from 1.0 to 1.5 in order to utilize the VSS logic rather than the VHS logic. The increase in the computation time was 12%, which is slightly greater than that due to running the data with the more general mixture program. The acceptance rate of collision pairs in the NTC collision routine was approximately 55% for this simple gas case.

The DSMC procedures are more readily understood at the level of the source code in the simple gas implementation DSMC0S.FOR, rather than in the more general DSMC0.FOR for the gas mixture.

11.3 Rotational relaxation and equilibrium

The program DSMC0R.FOR is an extension of program DSMC0.FOR and introduces an implementation of the Larsen–Borgnakke model for the selection of the appropriate rotational energies of diatomic or polyatomic molecules.

The rotational energy of molecule N is stored as PR(N), and this variable constitutes the named COMMON /MOLSR/. The named COMMON /GASR/ contains the information on the number of rotational degrees of freedom of each gas species and on the relaxation time. There is no reliable and universal theory for the variation of the rotational relaxation rate with relative speed in the collision or with the macroscopic temperature, and provision has been made for the rotational collision number to be specified as a second-order polynomial in the temperature.

The main test case data in subroutine DATA0R is for a simple diatomic gas with a constant rotational relaxation collision number of 5.0. The number of simulated molecules is 100,000 and these are in a single cell and sub-cell. Initially, the translational temperature of the gas is set to 500. and the rotational temperature to zero. The time step is set to about half the mean collision time, and the flow is sampled and the results are output at each step. The gas is assumed to come to equilibrium after 100 time steps or about 50 collisions per molecule. During this relaxation stage, the samples are set to zero at each time step, so that the macroscopic properties represent an instantaneous sample, and a time history of the temperatures is sent to the file RELAX.OUT. After the establishment of steady or equilibrium state, the sample of the macroscopic properties is time averaged.

The Larsen–Borgnakke calculation is in the subroutine INELR which is called in the collision subroutine between the SELECT and ELASTIC subroutines. The first step is to determine whether the rotational energy of either or both of the molecules is to be adjusted. The probability with which the rotational energy of a molecule is to be changed is equal to the reciprocal of the relaxation collision number Λ. This is used to set the indicator IRL to 0 if the rotational energy of molecule L is regarded as 'inelastic' and subject to change, or to 1 if it is 'elastic' and is not to be changed. The indicator IRM for molecule M is set in a similar manner. The variable ECC is set to the total energy that is to be divided, and this will be referred to as the 'available energy' for the Larsen–Borgnakke redistribution of energy. This loop over the two molecules also sets XIB as the total number of modes (translational and rotational) that participate in this redistribution (this corresponds to the parameter Ξ in §5.5).

The rotational energy of molecule L is first selected as long as IRL is equal to 1, and, secondly, that of molecule M is selected if IRM is equal to 1. At each selection, the mode that corresponds to that under selection is subtracted from XIB, so that this parameter corresponds to Ξ_b in the general Larsen–Borgnakke method. The ratio of the new rotational energy

to ECC is set as the variable ERM either through eqn (5.46) for two internal degrees of freedom, or the subroutine LBS for three degrees of freedom. The selection in the subroutine LBS is based on the general Larsen–Borgnakke distribution of eqn (5.43). The sum of E_a and E_b can be set to the available energy and is designated by E_c. The ratio of the probability of a particular value of E_a to the maximum probability can then be written

$$\frac{P}{P_{max}} = \left\{\frac{\Xi_a+\Xi_b-2}{\Xi_a-1}\left(\frac{E_a}{E_c}\right)\right\}^{\Xi_a-1}\left\{\frac{\Xi_a+\Xi_b-2}{\Xi_b-1}\left(1-\frac{E_a}{E_c}\right)\right\}^{\Xi_b-1}. \qquad (11.9)$$

The variables XIA and XIB represent Ξ_a and Ξ_b, respectively. The selection is based on the acceptance-rejection method, and the subroutine distinguishes between the general expression and the simplified expressions when either or both of Ξ_a and Ξ_b are unity.

In the case that both molecules are subject to selection, the available energy is reduced by the new rotational energy of molecule L, after the selection of that quantity. However, the initial or total available energy is accessible equally to molecules L and M. This is because the new rotational energy in molecule L could well be zero, and its initial rotational energy had been included in ECC. The end result is the same as if the total available energy was first divided between the translational and internal energies, and the internal energy was then divided between the molecules. The initial selection to a single internal mode has been employed because this procedure must be adopted for the quantized vibration procedure that will be developed in the following section.

The sampling routine is extended to cover the rotational energies, and the output includes the rotational temperatures of the mixture and the individual molecular species. The rotational temperature of species p is, from eqn (1.31),

$$T_{rot,p} = (2/k)(\overline{\varepsilon_{rot,p}}/\zeta_p), \qquad (11.10)$$

where ε_{rot} is the rotational energy of an individual molecule. The rotational temperature of the gas mixture is defined by

$$T_{rot} = (2/k)(\overline{\varepsilon_{rot}}/\bar{\zeta}), \qquad (11.11)$$

while the overall temperature is, from eqn (1.32),

$$T = (3T_{tr} + \bar{\zeta}T_{rot})/(3+\bar{\zeta}). \qquad (11.12)$$

The distribution functions of both the molecular speed and the rotational energies are also sampled and are output to the file DSMC0R.OUT.

The application of eqn (11.12) to the main test case data gives a nominal overall temperature of 300 K. Given the finite sample of 100,000, the actual overall temperature was 299.736 K. The time-averaged sampling of the temperatures at intervals of approximately half a mean molecular collision time, from 50 to 500 mean collision times, leads to a sampled

rotational temperature of 299.700 K and a translational temperature of 299.760 K. The sample size is 90,000,000 and the discrepancy is one part in 10,000. This corresponds to approximately one standard deviation, so that the procedures have been shown to lead to equipartition. Sample sizes of this magnitude require double precision arithmetic on 32 bit machines.

The approximate relaxation equation (5.68) was derived in §5.7 from an essentially macroscopic viewpoint that employed the internal and equilibrium (or overall) temperatures of the gas and did not take into account the finite sample of simulated molecules. When a molecule is accepted for rotational energy selection, the energy is chosen from the total collision energy, the average value of which is related to the overall temperature throughout the calculation. The overall temperature is a constant in this adiabatic relaxation process in a stationary gas, and another approximation to the relaxation equation can be derived from the Larsen–Borgnakke selection procedures. From the microscopic point of view, the relaxation can be regarded as a step process with the transition to equilibrium being completed when all molecules have been selected for rotational adjustment. The average energy of the fraction F_s of molecules that have been accepted for selection *one or more* times is directly proportional to the overall or equilibrium temperature T, while that of the fraction F_u that have not been selected is directly proportional to the initial rotational temperature $T_{rot,0}$. The rotational temperature can therefore be written

$$T_{rot} = F_s T + F_u T_{rot,0}. \tag{11.13}$$

The number of selections from a total of N molecules after time t is equal to $N\nu\Lambda t$, where ν is the collision rate per molecule and Λ is the fraction of inelastic collisions for rotation. Therefore, using the standard result from probability theory for the unselected fraction and noting that the sum of the fractions is unity, eqn (11.13) can be written

$$T_{rot} = T - (T - T_{rot,0})(1 - 1/N)^{N\nu\Lambda t}. \tag{11.14}$$

For $N >> 1$ and $N\nu\Lambda t >> 1$, eqn (11.14) can be approximated by

$$T_{rot} = T - (T - T_{rot,0}) \exp(-\nu\Lambda t). \tag{11.15}$$

in agreement with eqn (5.64).

For the main test case, $T = 300$, $\Lambda = 1/5$ and $T_{rot,0} = 0$, so that the prediction for the rotational temperature is

$$T_{rot} = 300\{1 - \exp(-\nu t/5)\}. \tag{11.16}$$

The translational temperature is given similarly by

$$T_{tr} = 300 + 200\exp(-\nu t/5). \tag{11.17}$$

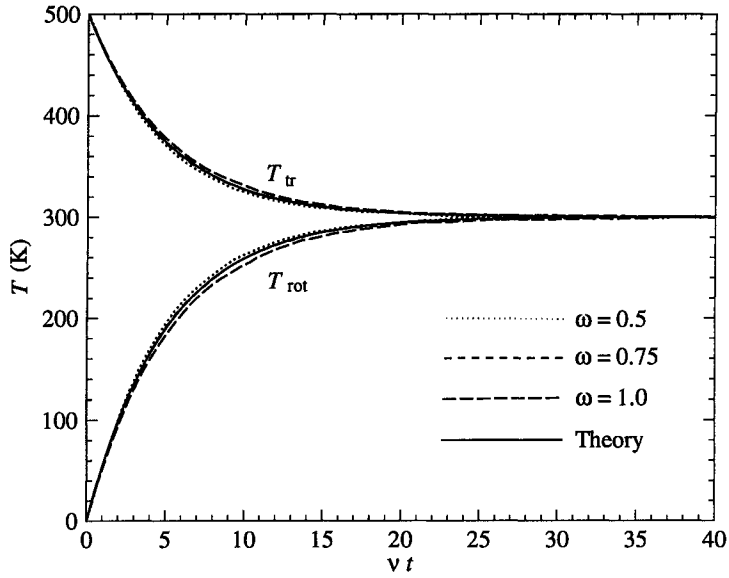


Fig. 11.2 Rotational relaxation in a homogeneous simple gas.

The DSMC results for the relaxation process are compared with the predictions of eqns (11.16) and (11.17) in Fig. 11.2. The theory is approximate and the dependence of the relaxation process on the viscosity–temperature index ω is in agreement with the more exact equation (5.70). It is fortuitous that the DSMC results for the realistic values of $\omega = 0.75$ and $Z_r = 5$ agree almost exactly with the simple approximate result. Figure 11.2 verifies that the Larsen–Borknakke relaxation as programmed in subroutine INELR is an exponential process when it is expressed on the 'collision time-scale'. Moreover, the relaxation collision number is given approximately by the reciprocal of the fraction of inelastic collisions in the Larsen-Borgnakke model. It should be noted that the abscissa in Fig. 11.2 is based on the sampled collision rate *per molecule* of the molecules. It is linear with time only in the 'Maxwell' case with $\omega = 1$, because the collision rate is then independent of the relative speed between the colliding molecules. Also, there will be a difference in the collision rate between the $\omega = 0.75$ case and the hard sphere cases. This creates problems in the application of measured relaxation times to the simulation of real gases. If the measured or 'handbook' values are quoted as 'collision numbers', it is essential to determine how the collision rate was established. There is a possibility of a factor-of-two error simply because the quoted relaxation collision number may be based on the rate of *collision events*, rather than *collisions per molecule*. In any case, the quoted rates will almost certainly be based on the unrealistic hard sphere model. A correction can be made

but, in the case of a temperature dependent relaxation rate, the temperature dependence of this rate must be reconciled with the temperature dependence of the collision rate that is present for any model other than the unrealistic Maxwell model. The VHS and VSS models have a definite cross-section, and the problem would be much more severe for models, such as the inverse power law model, that involve arbitrary cut-offs.

The situation is further complicated by the existence of alternative definitions of the relaxation rate. The preceding discussions have employed the conventional definition of eqn (5.67) that is based on the departure of the temperatures from the equilibrium value T_e. In this, the internal temperature is given by

$$\mathrm{d}T_i/\mathrm{d}t = (T_e - T_i)/\tau, \tag{11.18}$$

where τ is the relaxation time. Lumpkin et al. (1991) employed a similar definition, but with the translational temperature in place of the equilibrium temperature, i.e.

$$\mathrm{d}T_i/\mathrm{d}t = (T_{tr} - T_i)/\tau_a, \tag{11.19}$$

where τ_a is an alternative relaxation time. This definition is frequently expressed in terms of the energy rather than the temperature, and is employed in continuum flow studies when the equilibrium conditions are not known. The translational energy is then characteristic of a heat bath and the relaxation time is a function of pressure as well as temperature. Vincenti and Kruger (1965) note that this is not a relaxation process in the strict sense and call τ_a a *local* relaxation time. When eqn (11.19) is applicable and T_i is based on two internal degrees of freedom, the solutions of eqns (11.18) and (11.19) are similar and the alternative relaxation time is related to the conventional relaxation time by

$$\tau = 3\tau_a/5. \tag{11.20}$$

In addition to the difference in the definition of relaxation time, the results of Lumpkin et al. differ from those obtained from program DSMC0R as a consequence of a programming detail that may not, at first sight, be considered significant. In subroutine INELR, the Larsen–Borgnakke model is applied on a 'single molecule' basis in that each molecule is considered in turn for energy redistribution with probability Λ in each case. On the other hand, Lumpkin et al. apply this probability to the collision pair and, if a pair is accepted, the energy redistribution is applied to both molecules. The reason for the consequent difference in the relaxation rate can be seen from the theory that led to eqn (11.17). As long as most of the collisions that involve redistribution between the translational and rotational modes make the redistribution to only one of the molecules, a high proportion of the redistributions involve a molecule that has not previously been involved in a redistribution. The transfer is less effective if both of the molecules in a collision pair are always chosen for redistribution, and the program was temporarily modified to implement the 'collision pair' option.

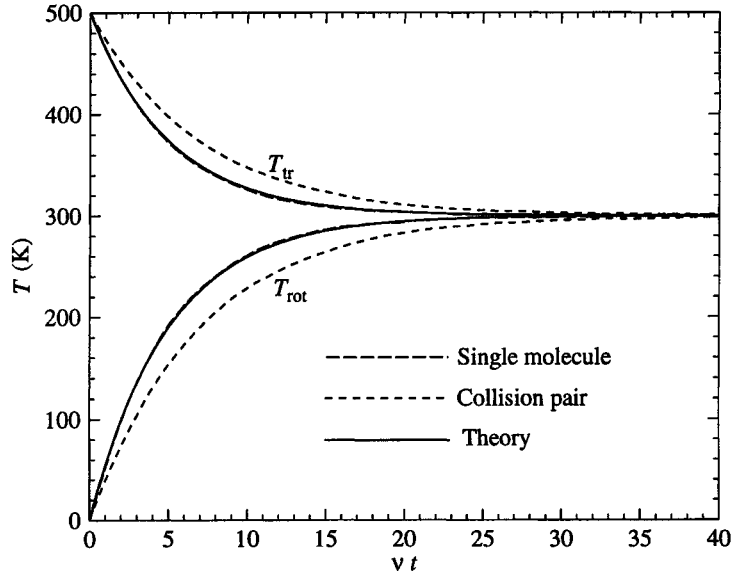


Fig. 11.3 The effect of the alternative selection procedures on the relaxation rate.

Either procedure can be used to model a given relaxation rate, but Fig. 11.3 shows that different values of Λ would be required. The program DSMC0R makes provision through the array SPR for the rotational relaxation probability of each species to vary with the species of the collision partner, and this can be implemented only when the 'single molecule' option is adopted. This example shows that the details of a procedure, at a level that would generally not be described in a journal publication, can lead to significant differences in the results and to consequent confusion. This case provides a salutory warning on the difficulties that can be caused by an inadequate description of DSMC procedures. Neither party may be aware that an alternative procedure can exist, let alone has been used.

Both the above discussion and the theory in Chapter 5 have reflected a preference for the relaxation collision number to be based on the relaxation time τ. However, the measured data for Z is usually based on the local relaxation time τ_a, and it must be adjusted before the Larsen–Borgnakke inelastic fraction can be set to $1/Z$ as a convenient approximation.

The subroutine INELR implements the Larsen–Borgnakke procedure in a hierarchical manner in that the energy is first divided between the translational and rotational components and, should both the molecules be chosen for redistribution, the rotational energy is then divided between the molecules. The 'single molecule' result in Fig. 11.3 has been obtained from the alternative subroutine INELRS that is also included in DSMC0R. This

implements the serial redistribution that was described in §5.5. Each molecule is considered for redistribution in turn and, should both be accepted, the redistribution to the second is with the translational energy that has already been modified by the first redistribution. It was shown in §5.5 that this leads to equipartition, and the result in Fig. 11.3 verifies that it does not affect the relaxation rate. The theoretical result is again based on eqns (11.16) and (11.17).

The molecular speed and rotational energy distribution functions are output to file DSMC0R.OUT at 400 points. The results for the speed distribution are compared in Fig. 11.4 with the theoretical distribution of eqn (4.6). While there is some scatter in the high-speed tail due to the small sample, it is clear that the collision procedures do not lead to any distortion of the Maxwellian distribution. In order to compare the rotational energy distribution with the equilibrium distribution, the constant in eqn (5.15) must be evaluated from the normalization condition that its integration from 0 to ∞ should be unity. The result is

$$f_{\varepsilon_r} = \frac{1}{\Gamma(\tfrac{1}{2}\zeta_r)\,kT}\left(\frac{\varepsilon_r}{kT}\right)^{\frac{1}{2}\zeta_r - 1} \exp\left(-\frac{\varepsilon_r}{kT}\right). \tag{11.21}$$

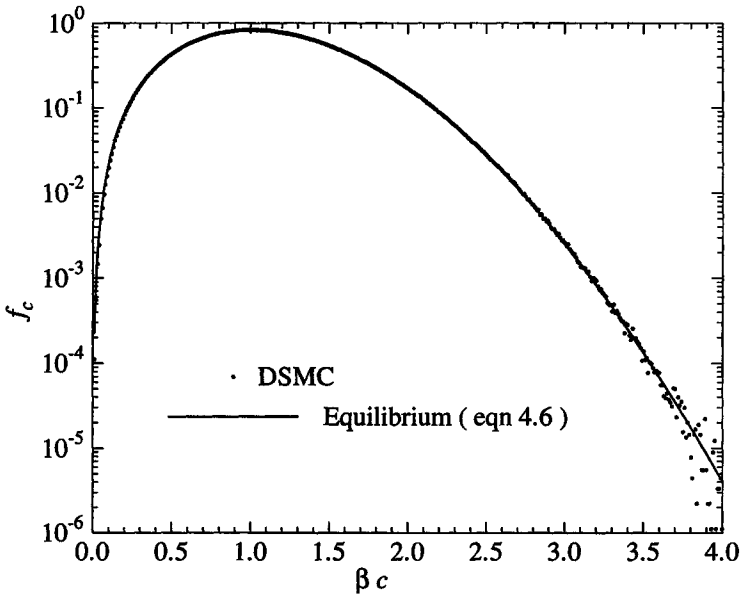


Fig. 11.4 The sampled distribution function for the molecular speed.

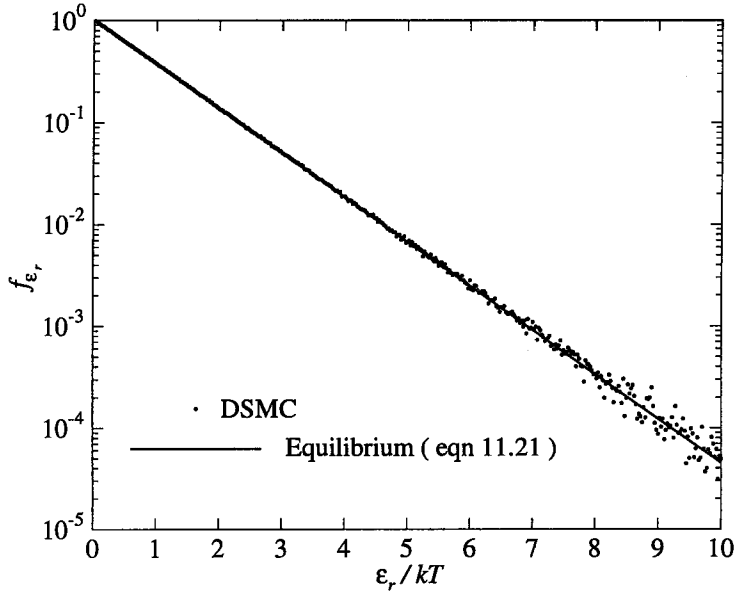


Fig. 11.5 Distribution function for the rotational energy of a diatomic gas.

For the special case of two rotational degrees of freedom, the maximum probability is for zero rotational energy and the probability decays as a simple exponential. This result is consistent with the Boltzmann distribution for the energy levels of a single internal mode. The main test case involves a gas with two degrees of freedom of rotation, and the sampled distribution is compared in Fig. 11.5 with the prediction of eqn (11.21). The two are in excellent agreement.

A similar rotational relaxation calculation was made for a gas mixture. The composition by number was 50% polyatomic molecules with three rotational degrees of freedom, 25% diatomic molecules with two rotational degrees of freedom, and 25% monatomic molecules with no internal degrees of freedom. The relaxation collision number was five for the dominant species and ten for the diatomic gas. The molecular masses were again set to nominal values.

The average number of internal degrees of freedom in this gas is two so that, for an initial translational temperature of 500 K, the final equilibrium temperature should again be 300 K. The computed temperatures of the mixture, based on a time average that was made after the establishment of equilibrium, were 299.74 K for the overall, 299.81 K for the translational, and 299.63 K for the rotational temperature. The translational and rotational temperatures for the polyatomic component were 299.77 K and 299.73 K, respectively. The similar values for the diatomic component

were 300.14 K and 299.32 K, while the translational temperature of the monatomic component was 299.55 K. This verifies that the definitions of the temperatures of both the gas mixture and the separate species, that were presented in eqns (11.10)–(11.12), lead to consistent values.

The relaxation behaviour was similar to that in the simple gas, but the collision rate per molecule is different for each species and, while the relaxation curves on the 'collision time-scale' conform to eqn (11.15), the absolute time-scale (i.e. in seconds) for the polyatomic gas differs from that in the diatomic gas.

The distribution of rotational energy in the polyatomic molecules is of interest. The most probable rotational energy of a molecule with ζ_r internal degrees of freedom is $(\frac{1}{2}\zeta_r - 1)kT$. This is zero for the diatomic gas, but is $\frac{1}{2}kT$ for the polyatomic gas. The distribution function from the DSMC computation is compared with the theoretical distribution function in Fig. 11.6. Note that these comparisons are to validate the coding in DSMC0R.FOR, rather than the Larsen–Borgnakke method itself. The method is based on the selection of the translational and internal energies from the relevant equilibrium distributions, so that any discrepancy would indicate an error in its implementation.

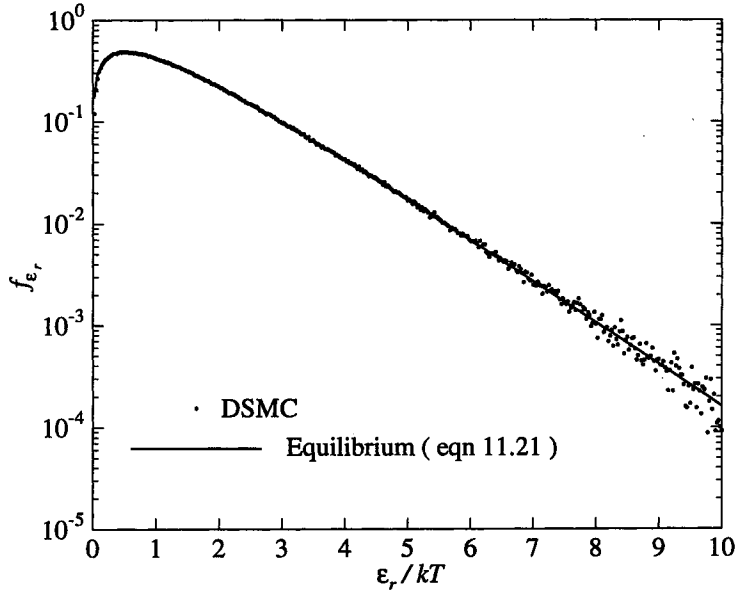


Fig. 11.6 Distribution function of the rotational energy in a polyatomic gas.

The Larsen-Borgnakke method is physically unreal in that only a fraction of the collisions are regarded as inelastic. Further criticism has been based on the selection of the post-collision values from an equilibrium distribution. This has led to claims that the method may be inadequate for highly non-equilibrium situations. However, it should be noted that the 'temperature' of the distribution changes from collision to collision, and does reflect the degree of non-equilibrium.

It was noted in §5.3 that a modified form of the Larsen-Borgnakke method had been introduced in respose to the first criticism. This has been implemented in the alternative subroutine INELRA in program DSMC0R. This regards all collisions as inelastic, but transfers only the fraction Λ of the calculated change in rotational energy at each collision. This did not affect the relaxation rate but, as had been predicted, the procedure does not satisfy detailed balance. The steady state rotational and translational temperatures were 297.83 K and 301.58 K, respectively. More seriously, the equilibrium distribution function of the rotational energy is drastically affected. Fig. 11.5 shows that the rotational energies are grouped in a narrow band around the mean rotational energy and that both high and low energies are under-represented. The modified model could therefore lead to errors in chemical reactions that depend on the rotational energy.

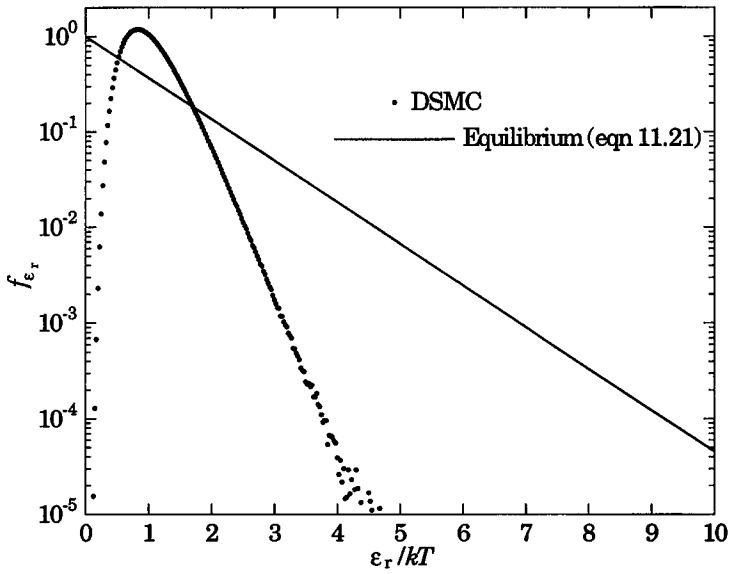


Fig. 11.7 The rotational distribution function for the modified Larsen–Borgnakke method.

11.4 Vibrational excitation

The program DSMC0V.FOR is concerned with the modelling of vibrational excitation. This program employs the harmonic oscillator model, and implements the quantum version of the Larsen–Borgnakke model that was outlined in §5.6. Not only is this more realistic than the classical version, but is more easily implemented and leads to a much faster execution time. The stored vibrational information on each molecule is the energy level of the vibrational mode. This program assumes that there is only one vibrational mode associated with each molecule, and separate levels would be required for polyatomic molecules that generally have more than one mode. The levels are relatively small integers and would be preferable to employ the short integers that are available with most compilers, but the program has been restricted to the FORTRAN 77 standard.

The data for the main test case is set in the subroutine DATA0V. It checks that a single vibrational mode in a simple diatomic gas remains in equilibrium with the translational and rotational modes and also samples the distribution of the levels. The initial temperature of the rotational mode in DSMC0R.FOR is always zero, but those of the rotational and vibrational modes in DSMC0V.FOR can be set to either zero or to the initial translational temperature FTMP. In the latter case, routines are required in subroutine INIT0V for the selection of representative rotational energies and vibrational levels that are appropriate to the temperature T.

First consider the selection of the rotational energy. For a diatomic molecule with two degrees of freedom, eqn (C16) allows the value to be directly related to a single random fraction R_f, i.e.

$$\varepsilon_r = -\ln(R_f)kT. \tag{11.22}$$

For ζ_r not equal to 2, the acceptance–rejection procedure may be used to find a typical value of ε_r/kT. The distribution function for this quantity is given by eqn (C13), and the ratio of the probability to the maximum probability is

$$\frac{P}{P_{max}} = \left(\frac{1}{\zeta/2-1}\,\frac{\varepsilon}{kT}\right)^{\zeta/2-1} \exp\left(\zeta/2-1-\frac{\varepsilon}{kT}\right). \tag{11.23}$$

The acceptance–rejection procedure for this ratio is implemented in the subroutine SIE.

Now consider the selection of a representative vibrational level of a molecule in a gas at temperature T. The harmonic oscillator model may be applied to a mode with characteristic vibrational temperature Θ_v. From eqn (5.57), the vibrational energy ε_v is limited to the discrete values given by $ik\Theta_v$, where the level i ranges from zero to infinity. There are two degrees of freedom associated with a vibrational mode, and the level may be determined from the truncation of the energy selected by eqn (11.22), i.e.

$$i = \lfloor -\ln(R_f)T/\Theta_v \rfloor. \tag{11.24}$$

The cumulative distribution function that led to eqn (11.22) can be written from eqn (C15) as

$$F_{\varepsilon_v/(kT)} = 1 - \exp\{-\varepsilon_v/(kT)\}. \tag{11.25}$$

The range of sampled energy $\varepsilon_v/(kT)$ that leads, after the truncation, to level i is from $i\Theta_v/T$ to $(i+1)\Theta_v/T$, and eqn (11.25) shows that the fraction of samples within this range is

$$\exp\{-i\Theta_v/T\} - \exp\{-(i+1)\Theta_v/T\}.$$

Therefore, the average value of the vibrational energy of a single molecule can be written

$$\begin{aligned}\bar{\varepsilon}_v = {} & k\Theta_v\{\exp(-\Theta_v/T) - \exp(-2\Theta_v/T)\} \\ & + 2k\Theta_v\{\exp(-2\Theta_v/T) - \exp(-3\Theta_v/T)\} \\ & + 3k\Theta_v\{\exp(-3\Theta_v/T) - \exp(-4\Theta_v/T)\} + \ldots\end{aligned}$$

or

$$\bar{\varepsilon}_v = k\Theta_v \exp(-\Theta_v/T)\{1 + \exp(-\Theta_v/T) + \exp(-\Theta_v/T)^2 + \exp(-\Theta_v/T)^3 + \ldots\}$$

This involves an infinite geometrical series with a multiplier less than unity, and the final result is

$$\bar{\varepsilon}_v = \frac{k\Theta_v}{\exp(\Theta_v/T) - 1}, \tag{11.26}$$

in exact agreement with the harmonic oscillator result of eqn (5.52). This result proves that the truncation to the quantum levels of the values selected from the continuous distribution for a fully excited mode properly accounts for the partial vibrational excitation of the gas.

The procedures for the vibrational excitation are integrated with those for the rotational excitation in the subroutine INELV. This program implements the serial application of the Larsen–Borgnakke method that was introduced in the final paragraph of §5.5 and on which the analysis of §5.6 was based. The molecules are considered in turn and the vibrational mode is considered before the rotational mode. The vibrational procedures are a direct implementation of the equations in §5.6, and the coding is related to the relevant equation numbers by comment statements.

The vibrational temperature is usually defined as the temperature in the harmonic oscillator result of eqn (5.52) that corresponds to the sampled value of the average vibrational energy. This can require an iterative process in the case of a gas mixture, and this definition cannot be applied to the more realistic anharmonic case. An alternative measure of the temperature is provided by the Boltzmann distribution of states. It can be seen from eqn (5.63) that, given the energy spacing between any two levels, the ratio of the number of molecules in the two levels can be used to define a vibrational temperature. In equilibrium, the temperatures based on all combiations of levels are equal and, for the harmonic oscillator model, the common temperature is equal to the temperature based on eqn (5.63).

Ideally, the temperature would be based on the slope of the line through all levels in a logarithmic plot of the level densities. However, to sample the number of molecules in a large number of levels in all cells would require excessive storage, and only two levels are sampled in this program.

The initialization and sampling routines include the array CSV which is a progressive count of the vibrational energy of each mode in each cell. This array is used in the output subroutine OUT0V to determine the average vibrational energy per molecule in each mode. The array CSVS is a progressive count of the number N_0 of molecules of each species in the ground state and the number N_1 in the first vibrational level. The vibrational temperature of species m can then be written

$$T_{v,m} = \Theta_{v,m} / \ln(N_{0,m}/N_{1,m}) . \tag{11.27}$$

The effective number of degrees of freedom in this mode is then

$$\zeta_{v,m} = 2\bar{\varepsilon}_{v,m}/(kT_{v,m}) . \tag{11.28}$$

Although this program employs equally spaced vibrational levels, these two equations do not depend in any way on the harmonic oscillator model.

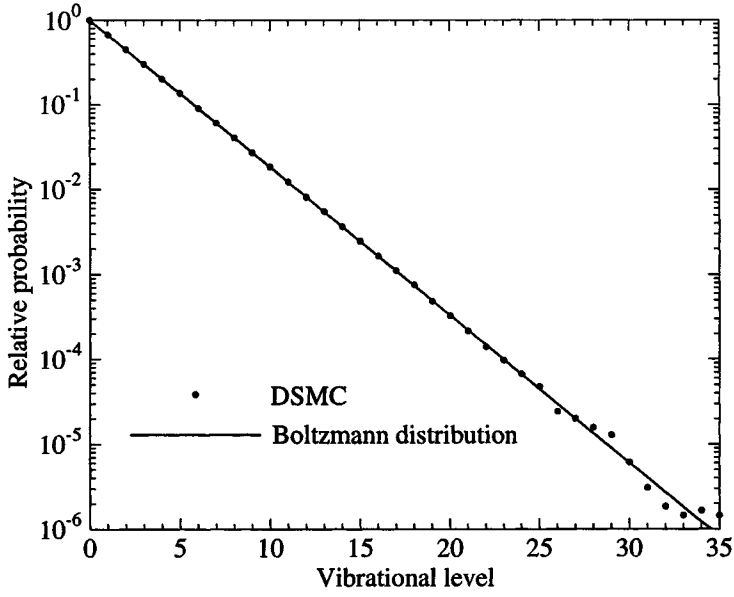


Fig. 11.8 The distribution function of the vibrational states in a gas with $\Theta_v = 2000$ K at a temperature of 5000 K.

The overall vibrational temperature is given by

$$T_v = \sum_{m=1}^{j} (\zeta_{v,m} T_{v,m}) / \sum_{m=1}^{j} (\zeta_{v,m}) . \tag{11.29}$$

The summation is over the modes for the temperature of a single species and over all modes of all species for the overall vibrational temperature. The overall temperature now becomes

$$T = (3T_{tr} + \bar{\zeta}_{rot} T_{rot} + \bar{\zeta}_v T_v)/(3 + \bar{\zeta}_{rot} + \bar{\zeta}_v). \tag{11.30}$$

The data in the listing of DSMC0V.FOR is for a diatomic gas which has a single vibrational mode with characteristic temperature 2000 K. The initial translational temperature was set to 5000 K and the 'run options' were chosen such that the rotational and vibrational temperatures were also set to this value. This run verified that the temperatures remained in equilibrium, and the time-averaged values for the overall, translational, rotational, and vibrational temperatures were 5006 K, 5007 K, 5009 K, and 5002 K, respectively. As shown in Fig. 11.8, the sampled vibrational distribution function is in excellent agreement with the Boltzmann distribution. This figure also shows that the temperature based on the first two levels is consistent with that based on the overall distribution.

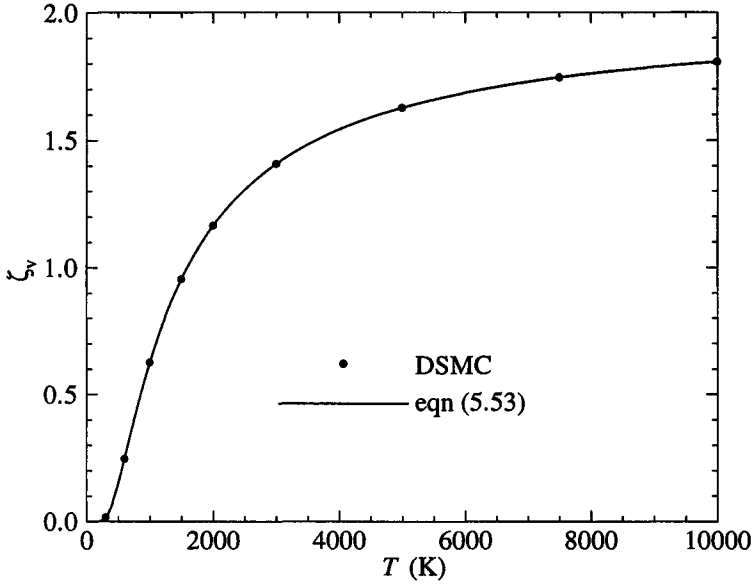


Fig. 11.9 Degree of excitation in a gas with $\Theta_v = 2{,}000$ K.

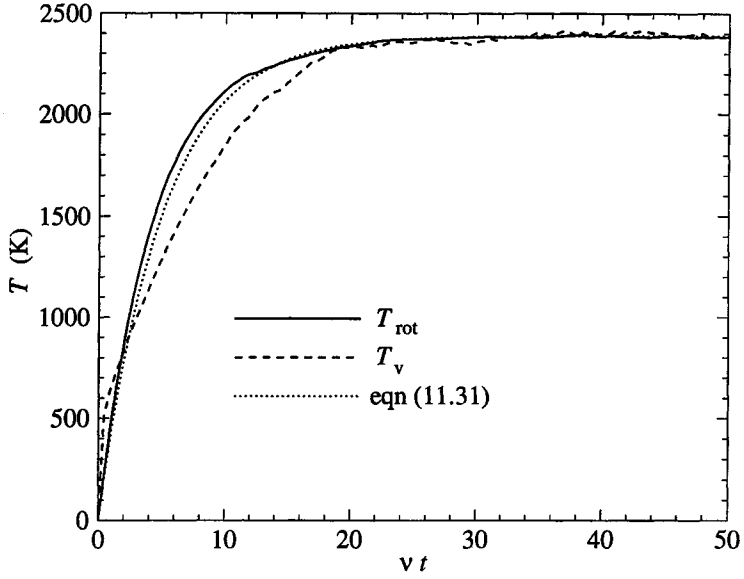


Fig. 11.10 Rotational and vibrational relaxation for Z_{rot}=5 and Z_v=5.

The effective number of degrees of freedom of vibration in the test case was found to be 1.630, and this is also in excellent agreement with the theoretical value of 1.627 from eqn (5.53). Similar runs were made for a number of other temperatures, and the degree of vibrational excitation as a function of temperature is compared in Fig. 11.9 with the prediction of the harmonic oscillator model. Again, the agreement is within the expected statistical scatter.

The listed test case was repeated with the initial rotational and vibrational temperatures set to zero and the initial translational temperature set to 5000 K. The final equilibrium temperature was then 2388 K and, if the quantum procedures were to have no effect on the relaxation process, the theoretical relaxation equation would be

$$T_v = T_{rot} = 2388\{1 - \exp(-\nu t/5)\}. \qquad (11.31)$$

The data on the relaxation process is written to the file RELAX.OUT, and the results for this case are shown in Fig. 11.10. The calculation employed 100,000 simulated molecules so that the expected level of statistical scatter is about 1%. The scatter in the rotational temperature is consistent with this, but, since the vibrational temperature is based on the number of molecules at the first level, the scatter is significantly higher. Allowing for the higher scatter, the relaxation behaviour of the vibrational temperature, as defined by eqn (11.27), is very satisfactory.

However, this definition cannot be used in situations where the degree of non-equilibrium is such that the distribution of molecules in the vibrational levels departs markedly from the Boltzmann form. For example, if there is a population inversion between the ground state and the first level, eqn (11.27) would lead to a negative temperature. In less extreme situations, such as the vibrational relaxation behind a very strong shock wave, the population of the upper levels increases more rapidly than that of the lower levels, and this leads to an unrealistically low vibrational temperature from eqn (11.27). More seriously, eqn (11.28) combines this low temperature with the actual mean value of the vibrational temperature and this can lead to values of the 'effective number of degrees of freedom' of a single vibrational mode well in excess of two. This is physically incorrect and, when substituted into eqns (11.29) and (11.30), it also leads to errors in the overall temperature. An alternative definition (Vincenti and Kruger 1965) may be based on the harmonic oscillator model of eqn (5.52). This is

$$T_{v,m} = \Theta_{v,m} / \ln(1 + k\Theta_{v,m} / \bar{\varepsilon}_{v,m}) . \tag{11.32}$$

If the quantum vibrational model is used, eqn (5.54) may be used to base this definition on the average level: i.e.

$$T_{v,m} = \Theta_{v,m} / \ln(1 + 1/\bar{i}) . \tag{11.33}$$

The effective number of degrees of freedom is then given directly from eqn (5.53) and eqns (11.29) and (11.30), for the overall vibrational temperature and temperature are still applicable.

The test case data employs constant rotational and vibrational relaxation collision numbers, but the program makes provision for these numbers to be functions of the temperature. The use of the macroscopic temperature ensures that the principle of detailed balance is satisfied, as noted in the discussion of the Larsen–Borgnakke model in §5.3, but the model is improved if the probability of energy redistribution is a function of a 'collision temperature' based in the energy in the collision. The Larsen–Borgnakke model contains an implicit temperature equal to the energy E_c that is being redistributed divided by the product of the effective number of energy modes that contribute to E_c and the Boltzmann constant. This temperature can be written

$$T_{coll} = E_c / (\Xi k), \tag{11.34}$$

where Ξ is given by eqn (5.42). The program DSMC0V.FOR employs the serial application option, and the Larsen-Borgnakke redistribution is always between the translational mode and a single internal mode. The parameter Ξ is therefore equal to $5/2+\omega_{12}$ plus either two for the rotational mode, or the equilibrium number of vibrational degrees of freedom for the vibrational mode. In order to test whether detailed balance can be satisfied with a collision temperature dependent relaxation rate, the option for a temperature dependent vibrational rate in DSMC0V.FOR employs the temperature T_{coll} that is defined in eqn (11.34). The contribution of ζ_v to Ξ is based on the variable SVIB that samples ζ directly from eqn (11.28).

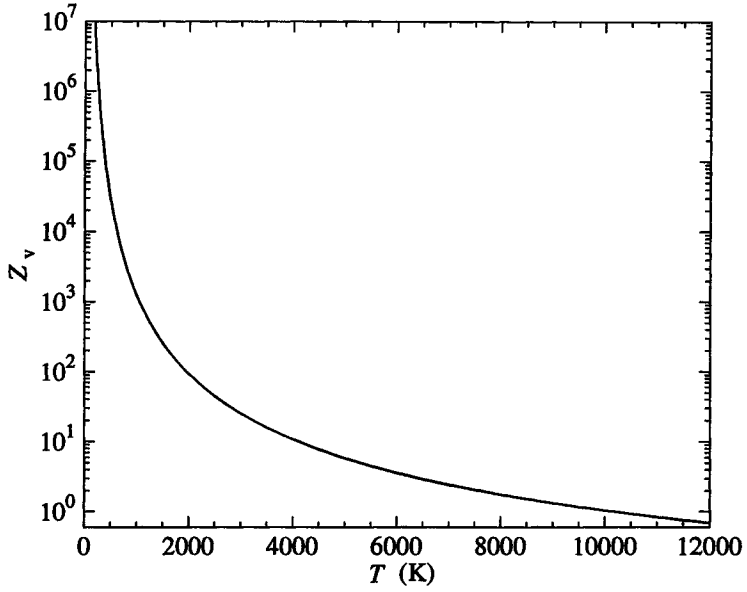


Fig. 11.11 Temperature dependence of vibrational relaxation rate.

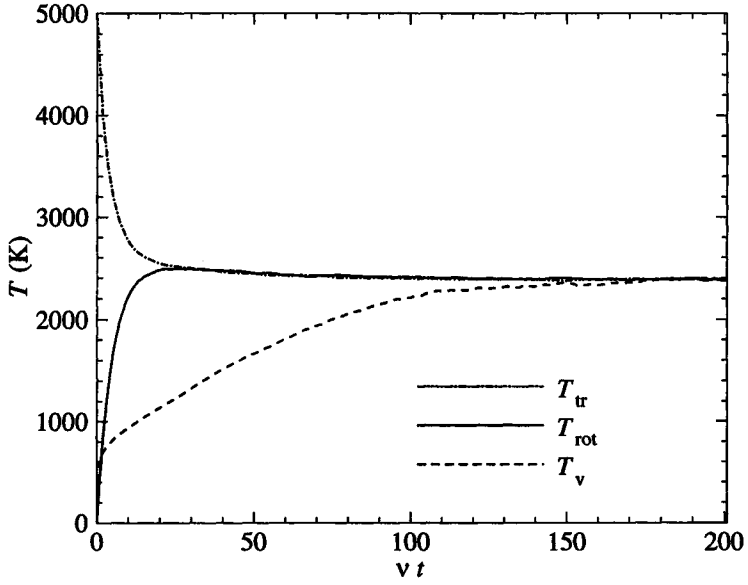


Fig. 11.12 Vibrational relaxation with temperature dependent rate.

The two test cases were repeated for a temperature dependent vibrational relaxation collision number with the constants C_1 and C_2 in eqn (6.53) equal to 10 and 100, respectively. The resulting vibrational relaxation collision number is shown as a function of temperature in Fig. 11.12. This is typical behaviour in that the collision number is extremely high at low temperatures, but drops below unity at high temperatures. The test case with all modes set to 5000 K confirmed that the 'collision temperature' dependent vibration rate is consistent with detailed balance. Both the average temperatures of the separate modes and the vibrational distribution function were indistinguishable from those for the corresponding case with a constant vibrational relaxation collision number. The results from the relaxation test case are shown in Fig. 11.12, and are more representative of a real gas. The rotational mode quickly comes to equilibrium with the translational mode, while the vibrational relaxation slows as the temperature drops. The time-averaged overall equilibrium temperature was 2387 K, while the translational, rotational, and vibrational temperatures were 2387 K, 2385 K, and 2386 K, respectively. More importantly, the distribution function for the vibrational energy was again in excellent agreement with the Boltzmann distribution.

11.5 Dissociation and recombination

The DSMC procedures for the modelling of real gases are extended to the dissociation and recombination reactions in the program DSMC0D.FOR. This program is limited to the dissociation of a single homonuclear diatomic gas, and nitrogen has been chosen for the test case. Dissociation is closely linked to vibrational excitation and the program employs the quantum model for vibration that was developed in §5.6. The vibrational procedure in this program is for the realistic anharmonic (or unequally spaced) levels.

The energy in vibrational level i of nitrogen is given by the following empirical equation (Herzberg 1950),

$$\varepsilon_{v,i} = k[3395\{(i+\tfrac{1}{2})-0.006126(i+\tfrac{1}{2})^2 + 0.00000318(i+\tfrac{1}{2})^3\}-1692.3]. \tag{11.35}$$

The zero-point energy has been subtracted in order to set the energy of the ground state (i=0) to zero. The characteristic temperature of dissociation is 113,200 K and this falls between levels 46 and 47. By contrast, the maximum level for the harmonic oscillator model would be 33. The amplitudes of the vibrational oscillations are often represented by the Morse potential which writes the potential energy of the oscillation as

$$U = E_d\{1-\exp(-\beta(r-r_e)\}^2. \tag{11.36}$$

This energy is zero when the internuclear distance r is equal to the distance r_e, and $r \to \infty$ as the potential energy approaches the dissociation energy E_d. For nitrogen, r_e is 1.094×10^{-10} m and the parameter β is 2.67×10^{10} m^{-1}. The Morse potential with these values has been used to construct Fig. 11.13. For a harmonic oscillator, the mean interatomic spacing is equal to r_e for all vibrational levels but, in the anharmonic case, Fig. 11.13 shows that it increases by up to 60% for molecules in the higher vibrational levels. This increase could be used to make an estimate of the increase in the collision cross-section for these molecules.

The energy in the levels is entered in the array ELEV as part of the data in the subroutine DATA0D. The characteristic vibrational temperature is set in the array SPD, as in DSMC0V.FOR, and the maximum level (just below the dissociation limit, is set in the array MLEV. Other data that relates directly to the dissociation–recombination reaction is the characteristic temperature of dissociation in SPD, and the information on the species codes in IDISS, IRECC, and IRSC.

A molecule dissociates when the Larsen–Borgnakke selection of energy into the vibrational mode leads to a level above the maximum level, and the dissociation rate should be consistent with the theory of §6.6. The 'equilibrium collision' theory for recombination is employed, and the ratio of the recombination cross-section to the elastic cross-section in an atom–atom collision is given by eqn (6.49).

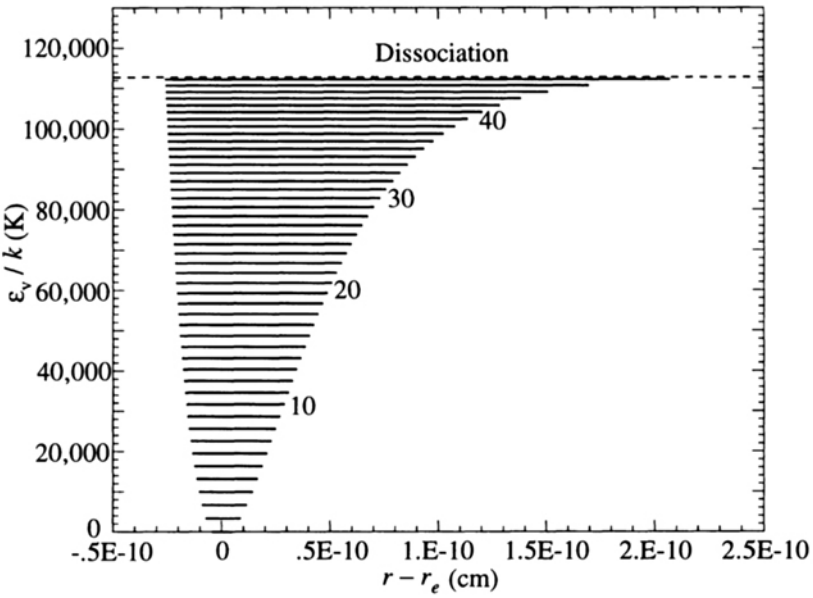


Fig. 11.13 The vibrational energy level diagram for nitrogen.

The Larsen–Borgnakke redistribution of energy in the collisions follows program DSMC0V.FOR and employs the 'serial' procedure in which each active internal mode interacts separately with the translational mode. This is modified for the consideration of dissociation in that the pre-collision rotational as well as the vibrational and translational energy is distributed between the vibrational and translational modes. Should the collision energy exceed the dissociation energy, a redistribution is made in order to determine whether dissociation occurs. The selection of possible levels for the acceptance–rejection scheme must extend beyond the dissociation energy and, for this purpose, notional levels are assumed with an energy spacing equal to that between the levels that span the dissociation energy. If the selected level is beyond the dissociation limit, the molecule is marked for dissociation by setting its species code to the negative of its proper value. The dissociation energy is subtracted from the collision energy, but a small amount of energy is stored as rotational energy and is subsequently converted to relative translational energy between the atoms. If the molecule was changed immediately into two atoms, the number of molecules would change and the indexing of the molecules would be upset. The marked molecule remains available for further collisions, but no further changes to its internal energies are allowed. The coding for the dissociation is in the subroutine DISSOC which is called after all collisions have been completed. The subroutine ELASTIC is used to compute a 'collision' between the newly formed atoms. The centre of mass velocity in this collision is equal to the velocity of the molecule, while the post-collision relative velocity is such that the translational energy is equal to the energy that had been assigned to the rotational energy of the molecule. This procedure leads to the exact conservation of mass, momentum, and energy in the overall computation of the dissociation.

The serial application of the Larsen–Borgnakke method means that the DSMC procedure corresponds to the derivation of eqn (6.52) for the dissociation rate coefficient due to collisions with atoms. The differing results for the molecule and atom as collision partners might appear to conflict with the logic behind the serial option for the application of the Larsen–Borgnakke method. However, the difference occurs only for dissociating collisions and comes about because of the disappearance of the rotational mode of the molecule in these collisions. The test case data in the listing of the program DSMC0D.FOR is for pure nitrogen, initially with the translational, rotational, and vibrational modes in equilibrium at a temperature of 30,000 K. The initial number density n_i of the gas is equal to the standard number density, and dissociations and recombinations proceed as the gas comes to chemical equilibrium. The rate coefficient of eqn (6.52) can be combined with eqn (6.2) to give a theoretical result for the initial rate of creation of atomic nitrogen. This result is compared in Fig. 11.14 with the results from DSMC0D.FOR.

The computed initial rate of formation of atomic nitrogen is about 25% higher than the theoretical rate. This is in spite of the theory being based

on the assumption that the vibrational collision number is unity, while the actual values of Z_v in the collisions with sufficient energy for dissociation in the simulation reduce the reaction rate by about one third. The theory differs from the simulation in that it employs a continuously distributed vibrational energy that necessitates the use of the harmonic oscillator model. The possible effects of the anharmonic model were tested by altering the data on the energy levels in DSMC0D.FOR to the uniformly spaced levels that correspond to the harmonic oscillator model. The initial rate of formation of atomic nitrogen was then 60% of the value for the anharmonic model. This resolves the discrepancy between the theory and the computation.

The reason for the higher reaction rate for the anharmonic model lies in the *uniform* choice of the possible levels prior to the application of the acceptance–rejection routine to eqn (5.61). Because the anharmonic levels are concentrated at the higher energy levels, this increases the probability of dissociation. It may be questioned whether the uniform choice of levels is correct. The effect may be avoided if the vibrational energy is chosen from the continuous distribution of eqn (C16) and then truncated to a level. However, if this is done, the equilibrium vibrational temperature is the order of 10% too low. The increased rate for the anharmonic model is therefore a real effect, and it brings the reaction rates from the 'exact available energy theory' of §6.6 into even closer agreement with the measured rates.

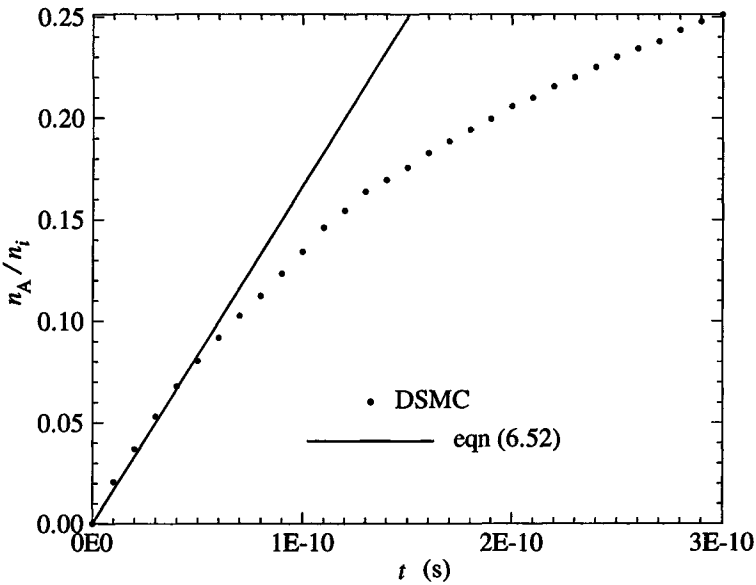


Fig. 11.14 Initial rate of formation of atomic nitrogen.

The recombination procedures in the program DSMC0D.FOR are based on the 'equilibrium collision theory' that led to eqn (6.35). The ratio of the partition functions that appears in this equation, and also in eqn (6.23) for the equilibrium degree of dissociation, can be written for a symmetrical diatomic gas as

$$\frac{(Q^{A})^2}{V Q^{A_2}} = \left(\frac{\pi m_A k}{h^2}\right)^{3/2} \Theta_r T^{1/2} \{1-\exp(-\Theta_v/T)\} \frac{(Q_{el}^{A})^2}{Q_{el}^{A_2}}. \qquad (11.37)$$

The values for nitrogen may be substituted into this equation and the result substituted into eqn (6.35), to give

$$\frac{\sigma_R}{\sigma_T} = \frac{9.907\times10^{-29}(1+41000/T)n}{T^{1/2}\{1-\exp(-3395/T)\}\{4+10\exp(-27658/T)+6\exp(-41495/T)\}^2}.$$

This is an excessively complex equation to evaluate at each collision and, to an accuracy of several percent over the temperature range of interest, it may be approximated by

$$\sigma_R/\sigma_T = 8\times10^{-27} n T^{-1}. \qquad (11.38)$$

The program assumes that the probability of recombination at each elastic collision is given by an equation of the form of (11.38), and the numerical coefficient and the temperature exponent are set in the array RECC. The factor C in eqn (6.49) has been set to unity and, because the forward rate is 25% higher than the theoretical rate on which the equation is based, the numerical factor was increased to 1.0×10^{26}.

At each collision between two atoms, the probability of recombination is calculated within the subroutine COLLMC and the cumulative probability PRC is increased accordingly. Each time that PRC exceeds unity, the two atoms are marked for recombination, PRC is reduced by unity, and the recombination switch IRECOM is changed from zero to unity. The marking in this case is by multiplying the species code by the parameter which denotes the maximum number of molecules and adding the address code of the atom with which it is to be recombined. Both the species code and the address of the recombination partner can be decoded from this integer. The actual recombination is coded in the subroutine RECOMB which, like DISSOC, is called after all the collisions for the time step have been calculated, and the indexing arrays are not required until they are reset at the next time step. Should IRECOM be unity, the molecules are scanned to determine a marked atom and the address of the recombination partner is decoded. The relative energy of these atoms is calculated, they are then combined to a molecule, and its velocity is set to the centre of mass velocity of the pair. The third-body molecule is then chosen by the subroutine SELECT, and the subroutine ELASTIC is then used to calculate an elastic collision between it and the new molecule. The energy in this collision is supplemented by the pre-recombination relative energy of the atoms and the dissociation energy. The Larsen–Borgnakke method is then applied to

redistribute this energy between the translational energy of the new molecule and third-body molecule, the rotational mode of the new molecule, and the vibrational energy of this molecule.

The gas composition in a run of DSMC0D.FOR changes until the rate of recombination equals the rate of dissociation. The gas is then in a state of chemical equilibrium. Eqns (11.37) and (6.23) may be combined with the physical data for nitrogen to give the following expression for the equilibrium degree of dissociation as a function of the initial number density n_i and the equilibrium temperature T.

$$\alpha^2/(1-\alpha) = (0.5047\times10^{28}/n_i)T^{1/2}\{1-\exp(-3395/T)\}\{4+10\exp(-27658/T) + 6\exp(-41495/T)\}^2\exp(-113200/T). \quad (11.39)$$

The electronic partition functions are included in eqn (11.37), even though electronic excitation is not included in the simulation. This does not lead to any inconsistency because the functions contribute equally to eqns (11.38) and (11.39), and the additional partition functions merely change the equilibrium state. The equilibrium degree of dissociation as a function of temperature is shown in Fig. 11.15 for the initial number density equal to the standard number density, and also for an initial number density of one tenth this value. Because three-body collisions are rare events in a dilute gas, the dissociation occurs at temperatures that are small in comparison with the characteristic temperature of dissociation.

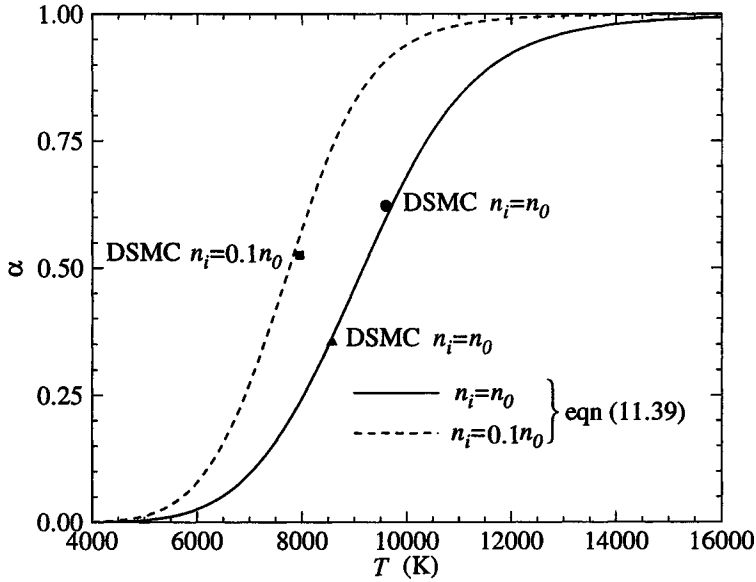


Fig. 11.15 Equilibrium degree of dissociation in nitrogen.

The listed test case with an initial temperature of 30,000 K relaxes to the temperature and degree of dissociation indicated by the circular symbol in Fig. 11.15. The effective number of vibrational degrees of freedom of the molecular nitrogen at equilibrium is 1.69, and this may be compared with the corresponding harmonic oscillator value of 1.67. The anharmonic effect on the overall energy in the vibrational mode is therefore very much smaller than the anharmonic effect on the dissociation rate.

If the initial temperature is reduced to 20,000 K, the gas relaxes to the conditions indicated by the triangular symbol in Fig. 11.15. The final test case was for an initial temperature of 25,000 K at a number density of one tenth the standard number density. The equilibrium state is indicated by the square symbol and, again, there is good agreement between the calculated and theoretical equilibrium states. In equilibrium at this lower density, there is approximately one recombination (and one dissociation) reaction for every million collisions. The fraction of recombinations decreases linearly with the gas density and, under rarefied conditions, recombinations are so rare that they can generally be ignored.

The integration of the dissociation model with that for the vibrational excitation has been shown to lead to satisfactory results for both the dissociation rate and the equilibrium state. As shown in §6.6, the exchange reactions can also be based on the vibrational relaxation rate. Carlson (private communication) has shown that the DSMC method can be used for five species real air calculations without there being any need for experimentally based data on the reaction rates.

11.6 Fluctuations and correlations

The effects of the statistical fluctuations are apparent in all results from the DSMC method. As discussed in §10.4, these fluctuations can be related to those in a real gas and, since they exist in an equilibrium gas that is macroscopically at rest, they are clearly non-dissipative. There are computational procedures that can cause the fluctuations to be enhanced or reduced as a result of numerical effects. Also, there are circumstances in which macroscopic fluctuations arise in a real gas, and these may be associated with changes in the nature of the microscopic fluctuations. It is therefore desirable to have a quantitative description of the fluctuations that occur in the DSMC simulation of a homogeneous gas.

The characteristics of the fluctuations have been determined by the program DSMC0F, which is based on the simple gas program DSMC0. The program samples the size distribution of the number fluctuations, the mean square number fluctuations of the macroscopic quantities, and the temporal and spatial correlation functions. The first step is to verify the assumption of §1.4 that the number fluctuations are consistent with the Poisson distribution of eqns (1.14) and (1.15). The data in the listed test case of DSMC0F is for an average nV of 100 molecules per cell. Fig. 11.16 shows that the sampled probability $P(N)$ of the number N during the run of DSMC0F is consistent with the prediction of eqn (1.15).

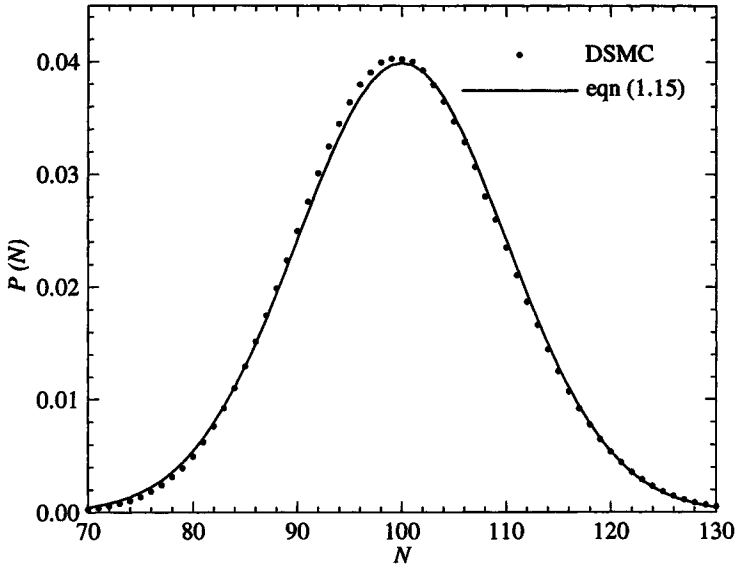


Fig. 11.16 The number distribution of simulated molecules in a cell.

The departure of the instantaneous number N from the average number nV is associated with a fluctuation δn in the number density. If the averages of macroscopic quantities are denoted by $\langle . \rangle$, the *mean square fluctuation* of the number density at a given location is defined by

$$K(0) \equiv \langle \delta n \, \delta n \rangle . \tag{11.40}$$

As long as the mean fluctuation is zero, the mean square fluctuation can be related to the mean and the mean square of the quantity. Again using the number density as an example,

$$\langle \delta n \, \delta n \rangle = \langle n \, n \rangle - \langle n \rangle \langle n \rangle . \tag{11.41}$$

This expression is employed in DSMC0F to calculate the mean square fluctuations of the number density and the velocity components in each cell. It cannot be applied to the temperature because, as can be seen from eqn (1.29), the fluctuations lead to spurious velocities in a stationary gas. These cause a systematic lowering of the temperature and, if this effect is regarded as contributing to the temperature fluctuation, the mean of this fluctuation is not zero.

The fractional fluctuation is of the order of the square root of the sample size, so that a normalized mean square fluctuation of the number density that would be expected to be of order unity is

$$\hat{K}(0) = (\langle \delta n \, \delta n \rangle / n^2) \langle N \rangle = \langle \delta N \, \delta N \rangle / \langle N \rangle . \tag{11.42}$$

The most probable molecular speed c'_m may be used to define the normalized mean square fluctuation of a velocity component, for example u, as

$$\hat{K}(0) = (\langle \delta u \, \delta u \rangle / c'^2_m) \langle N \rangle . \tag{11.43}$$

The sampled values of these normalized mean square fluctuations are shown in Fig. 11.17. There is considerable scatter in the mean square fluctuation of the number density, but the mean value across the flow is very close to unity. There is remarkably little scatter in the corresponding values for the velocity components, and these are all close to 0.5. Note that $c'^2_m = 2kT/m$, so that, if the normalization had been with regard to kT/m, the normalized values for the velocity components would also have been close to unity.

If the average is taken over the product of the fluctuations at two points that are separated in either distance or time we have the spatial and temporal *correlation functions*, respectively. For example, the normalized temporal correlation function of the number density is

$$\hat{K}(t) = (\langle \delta n_0 \, \delta n_t \rangle / n^2) \langle N \rangle = \langle \delta N_0 \, \delta N_t \rangle / \langle N \rangle . \tag{11.44}$$

The subscript $_0$ indicates the local fluctuation that was unsubcripted in the previous equations and the subscript $_t$ indicates the fluctuation at the same location, but separated by the time interval t. Note that the mean values that are employed in the normalization are not functions of time.

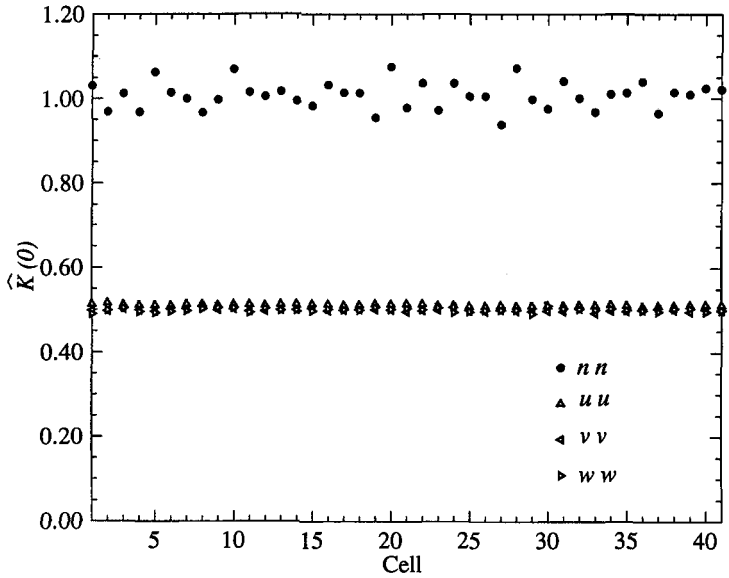


Fig. 11.17 The normalized mean square fluctuations in each cell.

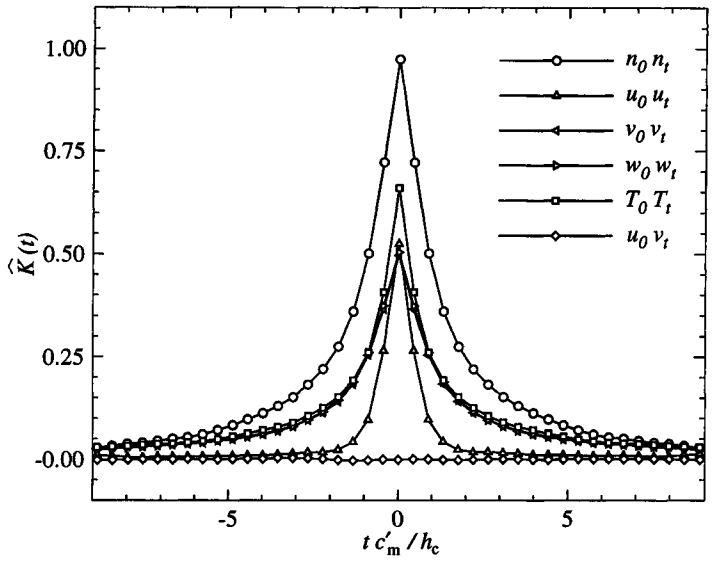


Fig. 11.18 The normalized temporal correlation functions.

The sampled time correlation functions for the cell midway between the boundaries in the flow of DSMC0F are shown in Fig. 11.18. This includes the temperature correlation function which is normalized in a similar manner to the velocity components, but with the mean temperature in place of the most probable molecular speed. The correlation function reduces to the mean square fluctuation as the time interval tends to zero. The u and v velocity components are uncorrelated even in this limit. The correlation function of the velocity component normal to the surfaces declines significantly faster than the other functions.

The time interval has been normalized by the cell height h_c divided by the most probable molecular speed. This is the time that a molecule moving with this speed normal to the cell boundary requires to cross the cell and this normalization takes no account of the mean free path and collision rate. The listed data in program DSMC0F is for a cell height equal to the one tenth the mean free path. The calculation was repeated with the collision routine disabled in order to obtain the collisionless or free-molecule result. It was also repeated for the density increased by a factor of five so that the cell height was half the mean free path. The three density correlation functions are compared in Fig. 11.19. A similar result is obtained for the other functions so that the density, and therefore the mean free path and collision rate, has very little effect on the decay of the correlation function. This is a somewhat surprising result in that the average diffusion rate of the molecules through the gas is strongly affected by the collision rate (Bird 1986).

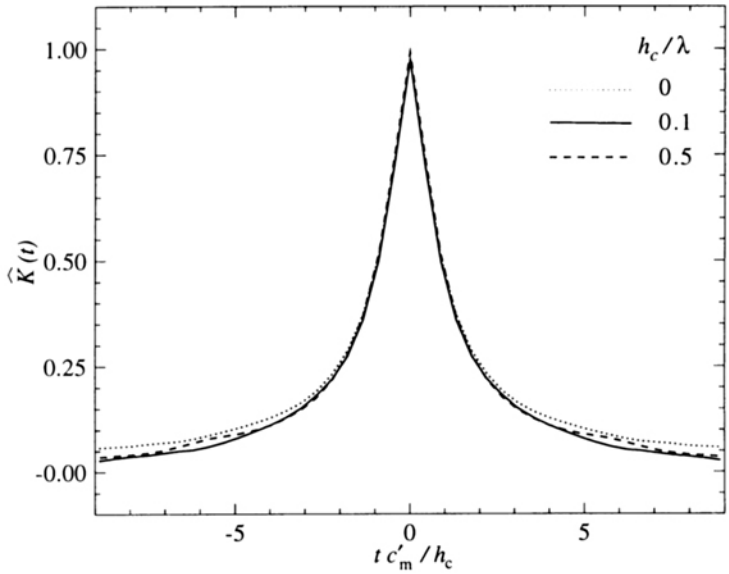


Fig. 11.19 The effect of density on the temporal correlation function.

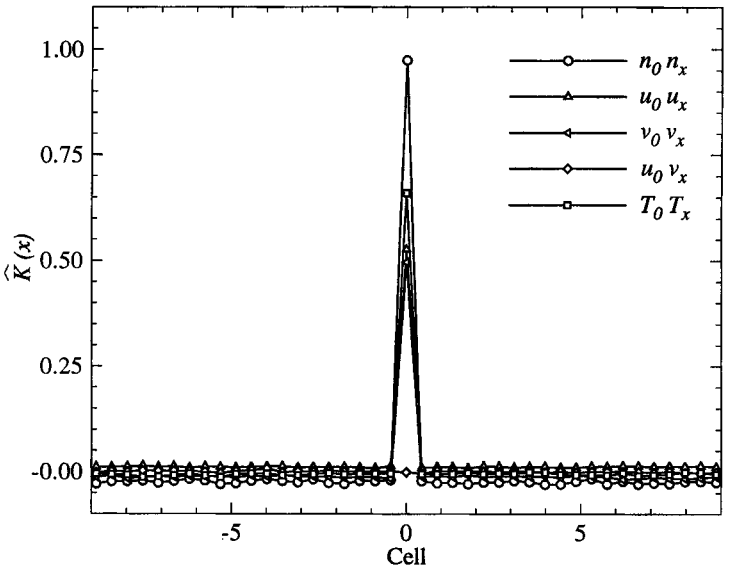


Fig. 11.20 The spatial correlation functions.

The normalized spatial correlation function of the number density in a macroscopically uniform gas is

$$\hat{K}(x) = (\langle \delta n_0 \, \delta n_x \rangle / n^2) \langle N \rangle = \langle \delta N_0 \, \delta N_x \rangle / \langle N \rangle . \qquad (11.45)$$

This function is again sampled in the central cell and this location is denoted by the subscript $_0$. The correlation function is plotted against the cell number, so that the spatial displacement x is effectively normalized by the cell height. The correlation function is again equal to the mean square fluctuation when the displacement is zero, but Fig. 11.20 shows that there is no sign of any correlation between the fluctuations in separate cells.

References

BIRD, G.A. (1976). *Molecular gas dynamics*, Oxford University Press.

BIRD, G.A. (1986). The diffusion of individual molecules within a gas. In *Rarefied gas dynamics* (ed V. Boffi and C. Cercignani), vol. 1, pp 400–409, B.G. Teubner, Stuttgart.

BIRD, G.A. (1989). Perception of numerical methods in rarefied gas dynamics. *Progr. Astro. and Aero.* **118**, 211–226.

HERZBERG, G. (1950). *Spectra of diatomic molecules*, Van Nostrand, New York.

KOURA, K. (1986). Null-collision technique in the direct simulation Monte Carlo technique, *Phys. Fluids* **29**, 3509–3511.

LUMPKIN, F.E., HAAS, B.L., and BOYD, I.D. (1991). Resolution of differences between collision number definitions in particle and continuum simulations, *Phys. Fluids* **3**, 2282–2284.

VINCENTI, W.G. and KRUGER, C.H. (1965). *Introduction to physical gas dynamics*, Wiley, New York.